Multi-Agent System: Theories and Applications

Lecture 5: Graphical and Mathematical Descriptions of MASs

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Outline

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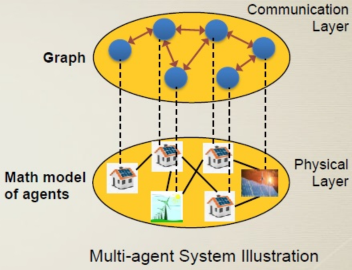
An MAS is a Cyber-Physical System



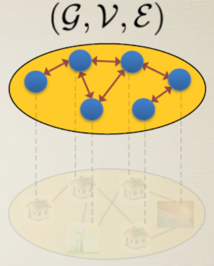
An MAS is a two-layer system

- **Physical layer:** where the real agents are located
- **Cyber layer:** where agents communicate with the others and are virtually represented in form of a *communication graph*

$\longrightarrow$ behaviors of an MAS depend on both dynamics of individual agents and inter-agent communication

MAS communication structure described by a graph

- each agent is represented by a vertex
- communication link between two agents is represented by an edge
- *graph could be undirected or directed depending on each specific MAS*
- *this communication graph does not depend on agents' dynamics, but influences agents' behaviors*



Behaviors of the whole MAS with digraph and undirected graph communications are different

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Linear MASs

$x_i(t) \in \mathbb{R}^{n_i}$: state, $y_i(t) \in \mathbb{R}^{p_i}$: output,
$u_i(t) \in \mathbb{R}^{m_i}$: control input, captures inter-agent communication

Continuous-time linear state-space model

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t)$$
$$y_i(t) = C_i x_i(t)$$

Discrete-time linear state-space model

$$x_i(t+1) = A_i x_i(t) + B_i u_i(t)$$
$$y_i(t) = C_i x_i(t)$$

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$$y_i(t) = C_i x_i(t)$$

Without control input: autonomous system

Continuous-time model

$$\dot{x}_i(t) = A_i x_i(t)$$
$$y_i(t) = C_i x_i(t)$$

Discrete-time model

$$x_i(t+1) = A_i x_i(t)$$
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Special cases

Continuous-time integrator

$$\dot{x}_i(t) = u_i(t)$$

mass point

Discrete-time integrator

$$x_i(t+1) = x_i(t) + u_i(t)$$

mass point

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Special cases

Continuous-time integrator

$$\dot{x}_i(t) = u_i(t)$$

mass point

Discrete-time integrator

$$x_i(t+1) = x_i(t) + u_i(t)$$

mass point

Continuous-time double integrator

$$\dot{x}_i(t) = v_i(t)$$
$$\dot{v}_i(t) = u_i(t)$$

vehicle-like dynamics

Discrete-time double integrator

$$x_i(t+1) = x_i(t) + v_i(t)$$
$$v_i(t+1) = v_i(t) + u_i(t)$$

vehicle-like dynamics

Homogeneity vs. Heterogeneity

An MAS is

homogeneous when $A_i = A_j \equiv A, B_i = B_j \equiv B, C_i = C_j \equiv C \ \forall \ i, j$. Otherwise, the MAS is called *heterogeneous* $\rightarrow$ more difficult to deal with.

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Mathematical model of the whole homogeneous MAS

$$\dot{x}(t) = (I_n \otimes A)x(t) + (I_n \otimes B)u(t)$$
$$y(t) = (I_n \otimes C)x(t)$$

$x(t) \triangleq \begin{bmatrix} x_1(t)^T & \cdots & x_n(t)^T \end{bmatrix}^T$, $u(t) \triangleq \begin{bmatrix} u_1(t)^T & \cdots & u_n(t)^T \end{bmatrix}^T$,
$y(t) \triangleq \begin{bmatrix} y_1(t)^T & \cdots & y_n(t)^T \end{bmatrix}^T$, $\otimes$: Kronecker product

Properties of Kronecker product

$$
\begin{aligned}
A \otimes B &\neq B \otimes A \\
A \otimes (B + C) &= A \otimes B + A \otimes C \\
(A + B) \otimes C &= A \otimes C + B \otimes C \\
(kA) \otimes B &= A \otimes (kB) = k(A \otimes B) \\
(A \otimes B)(C \otimes D) &= (AC) \otimes (BD) \\
(A \otimes B)^T &= A^T \otimes B^T \\
(A \otimes B)^{-1} &= A^{-1} \otimes B^{-1} \\
\det(A \otimes B) &= \det(A)^m \det(B)^n, A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{m \times m} \\
\mathrm{tr}(A \otimes B) &= \mathrm{tr}(A)\mathrm{tr}(B) \\
\lambda(A \otimes B) &= \lambda(A)\lambda(B)
\end{aligned}
$$

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Nonlinear MASs

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Continuous-time model

$$\dot{x}_i(t) = f_i(x_i(t), u_i(t))$$
$$y_i(t) = g_i(x_i(t), u_i(t))$$

Discrete-time model

$$x_i(t+1) = f_i(x_i(t), u_i(t))$$
$$y_i(t) = g_i(x_i(t), u_i(t))$$

$f_i : \mathbb{R}^{n_i} \times \mathbb{R}^{m_i} \mapsto \mathbb{R}^{n_i}$, $g_i : \mathbb{R}^{n_i} \times \mathbb{R}^{m_i} \mapsto \mathbb{R}^{p_i}$: nonlinear functions

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A lot of different nonlinear models, and in general, are more difficult than linear models